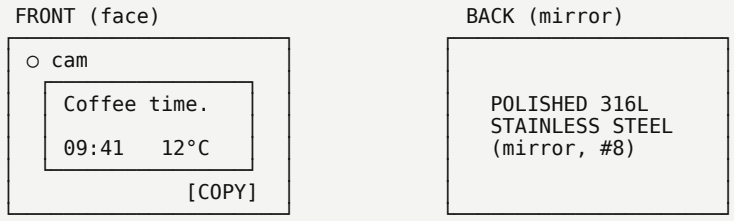
```
================================================================================
LOT-01 — HARDWARE PAGER FOR LOT-SYSTEMS.COM
PLAN · ARCHITECTURE · ROADMAP ANALYSIS
================================================================================
DOCUMENT    HW-00 / PLAN
ISSUE       2026.11.12 · REV A (session 01)
OWNER       Vadik — Inventor, COSMO® CIA
CLASS       INTERNAL / BUILD
STYLE       TERMINAL GRID
STATUS      CONCEPT → EVT PLANNING
================================================================================
```

00 · WHAT IT IS

LOT-01 is a 40 × 40 mm stainless-steel square. It does three things:

1. **Receives.** LOT-systems.com sends short notifications to it, the way a pager does: *"Coffee time."*, *"Rain at 16:00. Take the coat."*, *"Evening check-in."*
2. **Senses.** It has a weather sensor (temperature, humidity, pressure, air quality) and a small camera. These readings go back to the user's LOT account.
3. **Copies.** One button, **COPY**. Pressing it sends what is on the screen (and optionally a camera frame) to the user's **Log** tab on lot-systems.com.

Working name: **LOT-01**. The final product name is the inventor's call.



01 · THE 19 REQUIREMENTS, MAPPED

#	Requirement	How it is met	Doc
1	PCBWay	PCB fab, PCBA, stainless CNC and polish all ordered from PCBWay	HW-07
2	Pager-like notifications from AI site	LOT server writes ≤ 64-char messages; device pulls/gets pushed, buzzes	HW-05, HW-06
3	2-part stainless body	316L CNC: back cup (mirror) + front bezel frame, 4 × M1.2 screws	HW-03
4	Flat square 4 × 4 cm × 5 mm	40.0 × 40.0 mm; Z-stack budget in HW-03 §3 (5.0 target, 6.0 fallback)	HW-03
5	Camera	GC0308 VGA FPC module behind the cover glass (ESP32-S3 DVP)	HW-04
6	LOT API connector	`/api/device/*` connector, device tokens, spec + reference code	HW-06
7	PDF manuals	Every HW doc rendered to PDF in `docs/hardware/pdf/`	all
8	Compress info each session	Each session report ends with a COMPRESSION block (≤ 12 lines); device messages are compressed server-side to pager length	SR, HW-06
9	Firmware documents	HW-05 + source skeleton `hardware/firmware/lot01/` (main/main.c)	HW-05
10	Software to connect with firmware	Server connector `hardware/software/server/` + simulator	HW-06
11	Separate documents	One file per topic, HW-00 … HW-08	index
12	Charger	LOT Dock: Qi transmitter puck, USB-C powered	HW-04, HW-02
13	100 units run	Costing, PCBWay order plan, test jig, QA for 100 (+10 spares)	HW-02, HW-07
14	Weather sensor	Bosch BME688 (T / RH / P / gas) behind a vented membrane	HW-04

#	Requirement	How it is met	Doc
15	AI-grade off-the-shelf sensors	BME688 (on-sensor AI gas classes via BME AI-Studio), BMI270 IMU (gesture/wake), ESP32-S3 vector instructions for on-device ML	HW-04
16	COPY button → Log tab on lot-systems.com	`POST /api/device/copy` → new Log entry `event: device_copy`	HW-06
17	One side polished stainless	Back cup: 316L, mirror polish (#8), no openings	HW-03
18	Other side: camera, screen, button	Front: cover glass over screen + camera, steel COPY button in bezel	HW-03
19	Wireless charger	Qi receiver (TI BQ51050B) with coil **under the glass face**	HW-04

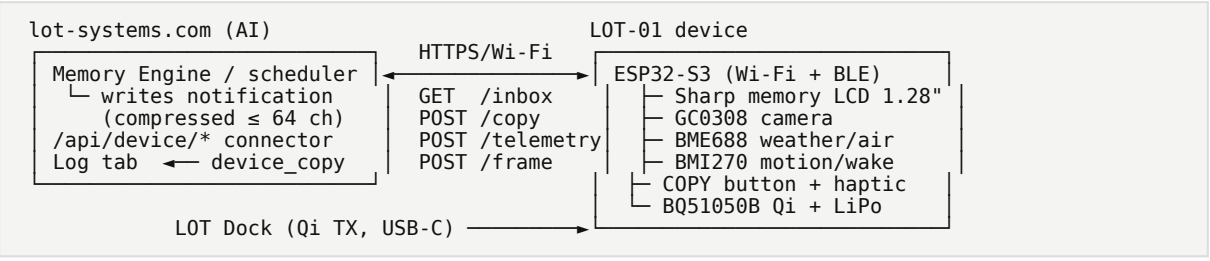
02 · THE THREE HARD PROBLEMS (read these first)

P1 — Steel blocks wireless charging. A Qi coil behind stainless steel does not charge; the steel heats up instead. *Decision:* the coil sits under the **front glass**, not the mirror back. The device charges **face-down** on the LOT Dock. The mirror back stays a clean, unbroken surface.

P2 — Steel blocks the radio. A closed steel box is a Faraday cage. Wi-Fi and Bluetooth must leave through the front glass and through a 0.3 mm non-conductive seam gasket between the two steel parts. *Decision:* chip antenna on the PCB edge, directly under the glass, with a keep-out zone. RF must be measured on the first EVT bodies, not assumed.

P3 — 5 mm is very thin. Glass + display + PCB + parts + battery + steel adds up to about 5.3–5.8 mm with real parts (HW-03 §3). *Decision:* EVT builds at **6.0 mm** to prove function. DVT pushes to **5.0 mm** using a 1.6 mm battery, a 0.4 mm PCB and a camera cut-out through the PCB.

03 · SYSTEM ARCHITECTURE



Data the device sends: COPY events, weather/air readings, optional camera frames. Data the device gets: notifications, time, config. Nothing else.

04 · ROADMAP

Phase	What	Exit gate	Weeks	Units
P0	Plan, BOM, docs (this session)	Docs signed off by inventor	1	0
P1	Dev-kit proto: ESP32-S3 dev board + breakout modules	COPY reaches Log tab; notification shows on LCD	2–3	3
P2	Server connector live on staging	Device token pairing works; simulator passes	1–2 (parallel to P1)	—
EVT	Custom PCB rev A (PCBWay) in 6 mm CNC body	Charges face-down; RF ≥ −70 dBm at 5 m; 3-day battery	4–6	10
DVT	PCB rev B, 5 mm body, polish finish, Dock	Drop 1 m ×10; 48 h soak; thermal on Qi ≤ 42 °C	4–6	20
PVT	100-unit run (+10 spares), test jig, packaging, manuals	100 % jig pass; yield ≥ 95 %	6–8	110
CERT	FCC/CE intentional radiator (pre-certified module helps)	Reports in hand before any sale	6–10 (overlaps PVT)	3–5

Total realistic calendar: **5–7 months** from today to 100 units in hand. Fastest path to a working object on a desk: **P1 in ~3 weeks**.

05 · ROADMAP ANALYSIS — RISKS AND CALLS

Risk	Prob.	Impact	Mitigation
RF through steel body too weak	High	High	Pre-certified module footprint as plan B (ESP32-S3-MINI-1 if Z allows); seam slot antenna; measure at EVT
5.0 mm not reachable with camera	High	Med	Ship at 5.5–6.0 mm, or make camera a DVT-optional feature
Qi heat on small coil (2.5 W into 90 mAh)	Med	Med	Cap charge at 0.5 C (≈ 45 mA); BQ51050B ILIM resistor; thermistor
Battery life with Wi-Fi	Med	High	Poll every 5 min in light sleep, memory LCD draws ~µW; target 3 days
Polished steel scratches during assembly	High	Low	Protective film from CNC shop to final pack; polish last
FCC/CE cost and time	Med	High	Use ESP32-S3 module with existing modular grant if thickness allows; budget US$ 8–15 k
BME688 reads hot (self-heating, ESP32)	High	Med	Isolate with PCB slot; vent membrane on bezel edge; software compensation
Camera privacy perception	Med	Med	Camera only fires on long-press COPY; LED/screen shows "CAM" when active

Recommendation: Start P1 (dev-kit prototype) and P2 (server connector) now, in parallel. Do not order EVT boards before P1 proves the COPY → Log loop end-to-end. Lock the 5 mm vs 6 mm decision at the end of EVT using measured parts, not datasheets.

06 · DOCUMENT SET

ID	File	Contents
HW-00	LOT-01-00-PLAN-AND-ROADMAP.md	This plan
HW-02	LOT-01-02-BOM-BUYING-LIST.md	Buying list, links, 100-unit costing
HW-03	LOT-01-03-MECHANICAL-ENCLOSURE.md	2-part steel body, Z-stack, CNC spec
HW-04	LOT-01-04-ELECTRICAL-DESIGN.md	Block diagram, pin map, power, Qi
HW-05	LOT-01-05-FIRMWARE.md	Firmware architecture and states
HW-06	LOT-01-06-LOT-API-CONNECTOR.md	Device ↔ lot-systems.com protocol
HW-07	LOT-01-07-MANUFACTURING-100-UNITS.md	PCBWay order, test jig, QA
HW-08	LOT-01-08-USER-MANUAL.md	Owner's manual
SR	sessions/LOT-HW-SR-YYYYMMDD-NN.md	Session report + compression block

PDF copies of every document: `docs/hardware/pdf/`.

07 · SOURCES AND LIMITS OF THIS REVISION

- brand.lot-systems.com, lot-systems.com/about and institute.lot-systems.com/cqgs.html **could not be read** in this session: the network policy blocks them. Brand language was taken from the repository (`docs/technical/LOT-STYLE-GUIDE.md`, LOT Wiki, which lists *"LOT® Station, LOT® Brush, COSMO® node"* as planned QIoT™ devices). Re-read those pages next session and align naming.
- Supplier sites were also blocked, so **prices are planning estimates** and links are manufacturer pages or distributor search links. Confirm price and stock before ordering.